Sumit Kumar Ranjan

Software Engineer | Backend, LLM & RAG Systems | Mohali, Punjab, India
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Summary

Software Engineer building backend services, event-driven pipelines and LLM applications. Hands-on with FastAPI microservices, Apache Kafka, PostgreSQL, Retrieval-Augmented Generation (RAG) and Text-to-SQL. Oracle Certified Generative AI Professional, co-author of a Springer-published research paper, and 630+ DSA problems solved.

Experience

Pharmaco Evidence **Aug 2025 – Present**
Software Engineer *Mohali, Punjab*

- Build and maintain backend microservices in **Python** and **FastAPI** that power AI-driven evidence products.
- Develop event streaming pipelines on **Apache Kafka** to move data between services asynchronously.
- Design and integrate **REST APIs** and third-party services, with data stored in **PostgreSQL**.

EOXS **May 2025 – Aug 2025**
AI Developer Intern *Remote*

- Built AI modules inside the EOXS ERP for automation and predictive analytics in the steel industry.
- Designed **machine learning** models for demand forecasting and operational efficiency.
- Shipped **Contract Review AI**, a PO–SO comparison tool (Next.js, React) that pulls Sales Order PDFs from **Odoo ERP** over XML-RPC/REST and flags discrepancies with an LLM, cutting manual review effort by **90%**.

Projects

Sonia – RAG Document Chatbot | GitHub **2025**
FastAPI, Next.js, Gemini, LlamaParse

- Built a RAG chatbot with layout-aware PDF parsing (**LlamaParse**) and **two-stage retrieval**: vector search for the top 25 chunks, then a **cross-encoder re-ranker** for the top 5.
- Added SHA-based duplicate detection against **ChromaDB** so known PDFs skip re-processing, reducing processing time by **80%**.

Claim AI Assistant – Text-to-SQL + RAG | GitHub **2025**
React, FastAPI, PostgreSQL, LangChain

- Users upload insurance claims as CSV and ask questions in plain English; a router sends structured queries to **Text-to-SQL** on PostgreSQL and reasoning-heavy ones to **RAG**.
- Streamed LLM answers to the React client for a responsive chat experience.

Kite – SaaS Delivery Marketplace | Live **2026**
React, FastAPI, PostgreSQL

- Built with a four-person team: a marketplace connecting clients with vetted developers, with milestone-based payments and invoices.
- Implemented **role-based access control** for clients, team leads and admins, plus task boards, project chat and status tracking.

RideBridge – Exact-Pickup Ride Linker | Live | GitHub **2026**
React, Vite, Google Maps & Places API

- Lets users pin the exact gate of large venues by search, voice input or tap, then passes exact coordinates into **Uber via deep links** and **Rapido via Android intents**.

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL

Backend: FastAPI, Flask, REST APIs, Microservices, Apache Kafka

AI / ML: LLMs, RAG, LangChain, LlamaIndex, LangSmith, Prompt Engineering, PyTorch, TensorFlow, scikit-learn, YOLOv8, OpenCV

Frontend: React, Next.js, Tailwind CSS, Three.js

Databases & Cloud: PostgreSQL, MySQL, SQLite, ChromaDB, AWS, Azure, Oracle Cloud (OCI), Linux, Git

Publications

- **Cascade-Level Fake News Classification Using Graph Attention Networks over Diffusion Trees** – Co-author, presented at **BIDA 2026 (Springer)**.

Education

Chandigarh University **2022 – 2026**

B.E. Computer Science and Engineering (AI & ML) – CGPA: 8.04 *Mohali, Punjab*

Certifications

- Oracle Cloud Infrastructure 2025 Certified **Generative AI Professional** – Oracle (2025)
- Microsoft Certified: **Azure AI Fundamentals (AI-900)** – Microsoft (2024)
- AWS Academy Graduate – Cloud Foundations – Amazon Web Services (2023)

Achievements

- **1st Runner-up, BIZINNOVATHON '24** (Bajaj Allianz) – national hackathon, 100+ teams; won INR 15,000 and internship offers.
- **8th rank of 700+** in Hash It Out, a national coding contest by Coding Ninjas (2024).
- **630+ DSA problems** solved across LeetCode, CodeChef, Code360, GeeksforGeeks and Codeforces.